Data → Train → pretrain model → Train on Specific Data → finetune

un-supervised training

SFT

Raw text train on → pretrain data

Label data train → finetune

finetuning

- Classification
- instruct fine tune

Lecture-04

Attentation all you Need

Transformers : deep nural network

Vector embedding

King
Queen
apple
banana

kpi's standard benchmark

Encoder

embed input text into vector.

Decoder

Generator output text from encoder vector

Self-Attentation: each sentence how much related to each other
↳ its allowed to long range dependency

GPT — Generative pretrain Transformer

Bert → predict hidden word in a given sentence
↳ mask Language model

GPT — predict the next word.

One word at a time

Bert
encoder

- Best for sentiment Analysis
- Sentence look both dir

GPT
Decoder

its left to right

Transformer

LLm

- Not all transformers are not llm
- Transformers can also be used for computer Vision.
- All llm are transformer.

Lecture-05

How GPT Works :

Zero-Shot : Ability to generate to completely unseen task without point specific example

Few Shot : Learning from a minimum of example which the user provides as inputs.

One-shot — One example only

⇒ GPT-3 Has use to train on data — public data

- Common crawl — 60
- WebText 2 — 22
- Books1 — 8
- Books2 — 8
- wikipedia — 3

3,00 billion data

Auto-regressive : previous output uss as input for next output prediction.

GPT — unsupervised and Auto-regressive

GPT
- Un-supervised
- Autoregressive

paper

Emergent Behavior :

GPT train for next word prediction but its can also Robust for other tas as classification, Translation instruct task its called emergent behaviour.

Lecture-06

Stage of building LLm :